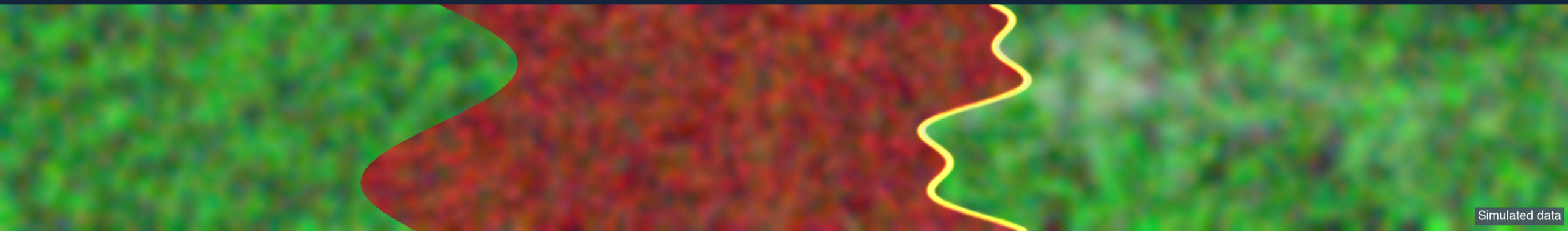
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

# Wildfire Monitoring with Remote Sensing

Finding fires, mapping burn scars and judging fire danger from space

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

# Use satellite data to detect fires as they burn, map what burned and how badly, and judge where the next fire is likely.

The physics of seeing heat, burn-severity mapping, fire danger, real fires from Uttarakhand to Los Angeles, and engineering use.

THIS LECTURE

# Outline


**01** **Wildfire and why we monitor it**
Fire behaviour, fires in India, and the before–during–after cycle

**02** **Detecting active fires**
Seeing heat, sub-pixel fires, satellites and fire radiative power

**03** **Mapping burned areas and what follows**
NBR, burn severity, radar, recovery and post-fire hazards

**04** **Before the fire: fire danger**
Dry vegetation, fire seasons and danger ratings

**05** **Case studies**
Uttarakhand, Punjab, Australia, Canada and Los Angeles

**06** **Engineering practice**
Uses, a workflow, limits and exercises

01

PART 1

# Wildfire and why we monitor it

Fire behaviour, fires in India, and the before–during–after cycle

WILDFIRE

# What drives a wildfire

**Fuel**

Grass, leaf litter, shrubs and trees. How much there is, and how dry it is.

**Terrain**

Fire runs uphill faster; slopes and valleys channel the wind.

**Weather**

Heat, low humidity and wind dry the fuel and push the fire forward.

**Ignition**

Mostly people in India: land clearing, carelessness, burning crop residue. Also lightning.

WILDFIRE

# Fires in India

**Forest fires**

Mostly surface fires in the dry season, February to June: chir pine forests of the Himalaya, and the dry deciduous forests of central and eastern India.

**Crop-residue burning**

Paddy stubble burned in Punjab and Haryana in October–November, a major source of winter smoke over Delhi.

**Satellite alerts**

The Forest Survey of India has sent fire alerts from MODIS since 2004 and from VIIRS since 2017, to every state forest department.

WILDFIRE

# Before, during and after

### Before

**Where is it dry enough to burn?**

Vegetation moisture, temperature, weather, fuel maps

### During

**Where is it burning now, and how intensely?**

Active-fire hot spots, fire radiative power, smoke

### After

**What burned, how badly, and what next?**

Burn scars, severity, recovery, erosion and debris-flow risk

WILDFIRE

# Why satellites for fires

**3 hours**
NASA FIRMS delivers fire detections worldwide within about 3 hours

**375 m**
Pixel size of VIIRS active-fire detections

**5–10 min**
Geostationary satellites image the same area again

**Since 2000**
Daily MODIS fire record for fire history

### Remote forests
Fires in hills and jungles are often seen first from space.

### Whole seasons
Burned area for every district, every year, to plan and budget.

### But not a ground crew
A satellite sees a hot pixel, not where exactly the flames are or how to reach them.

02

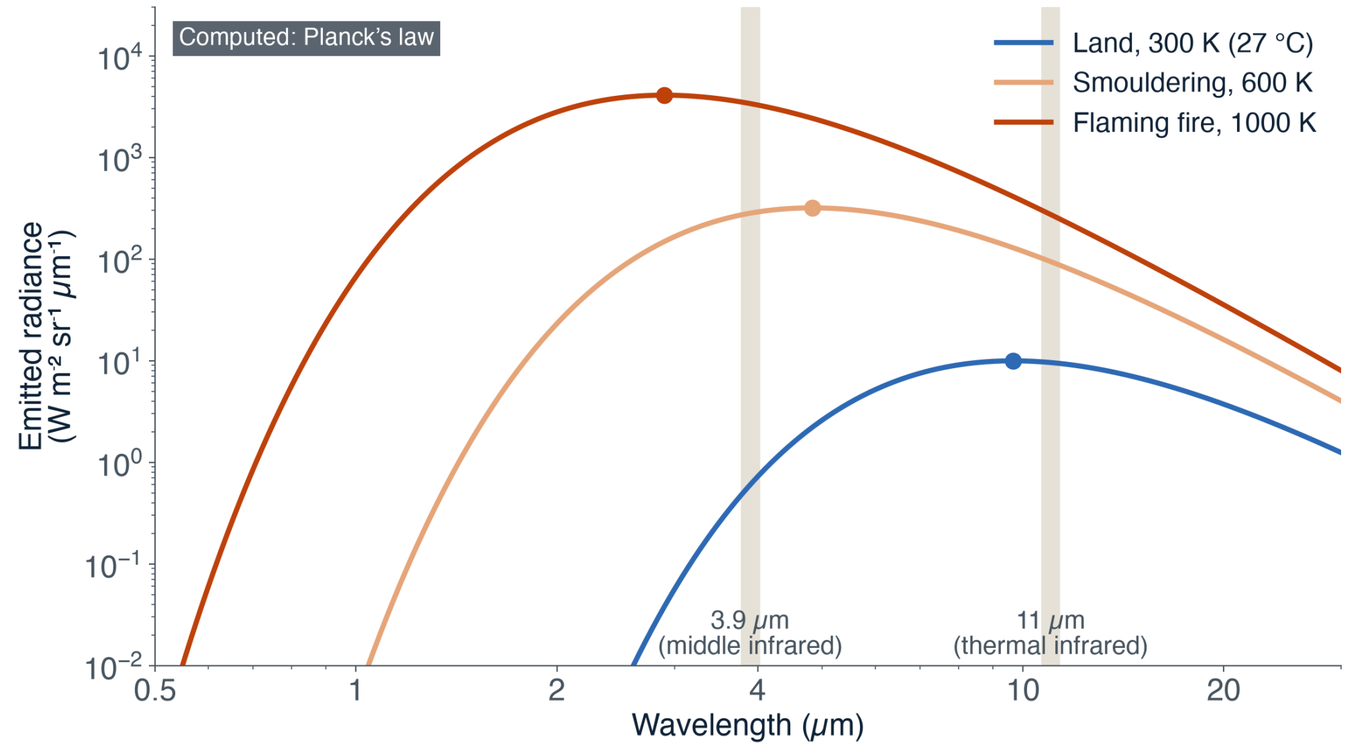
PART 2

# Detecting active fires

Seeing heat, sub-pixel fires, satellites and fire radiative power

ACTIVE FIRES

# Hot things glow at shorter wavelengths




Computed from Planck's law. Dots mark each peak.

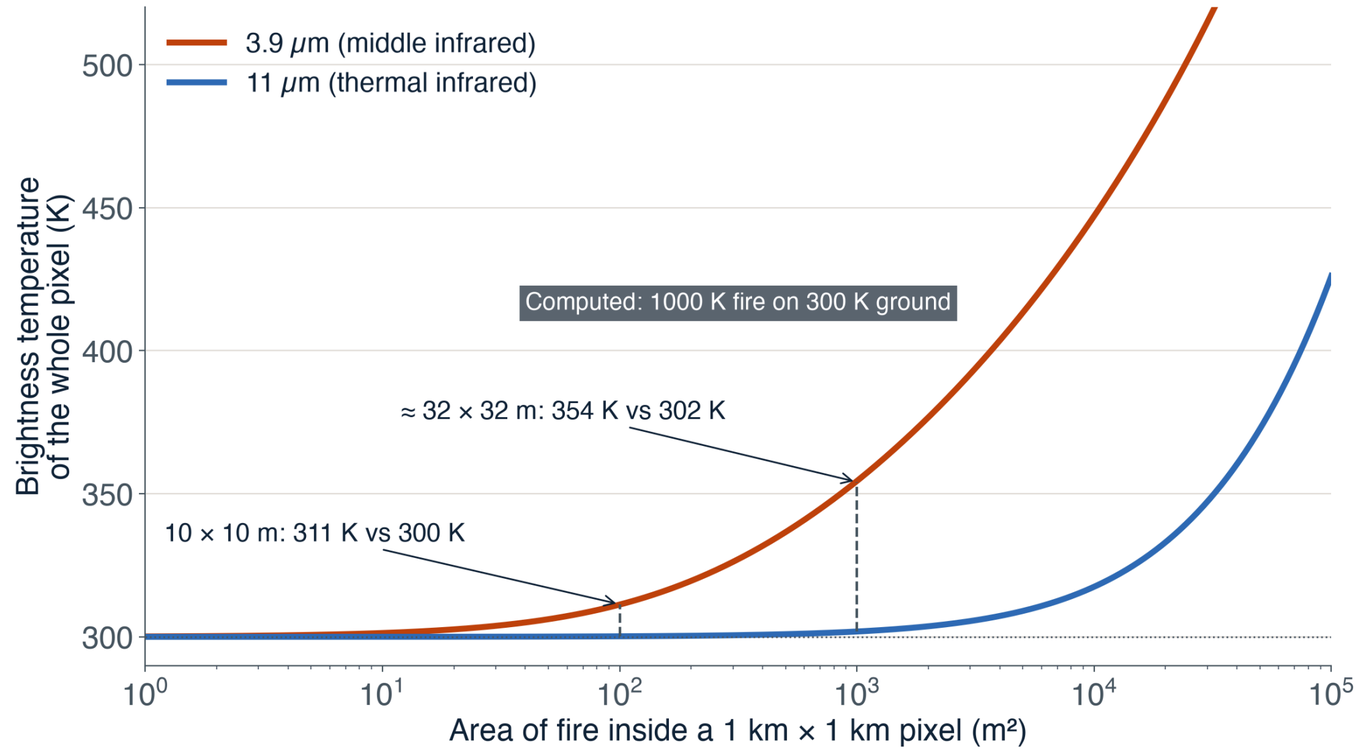
Ground at 300 K glows most at about 9.7 µm; a 1000 K fire at about 2.9 µm (Wien's law: λ = 2898 ÷ T).

At 3.9 µm the fire is about 5.6 thousand times brighter than the ground; at 11 µm only about 29 times.

So fire sensors compare a middle-infrared channel with a thermal one.

ACTIVE FIRES

# Small fires, big pixels




Computed: a 1000 K fire inside a 1 km pixel of 300 K ground.

A fire far smaller than the pixel still raises its 3.9 µm temperature.

10 × 10 m of fire: 311 K at 3.9 µm, 300 K at 11 µm.

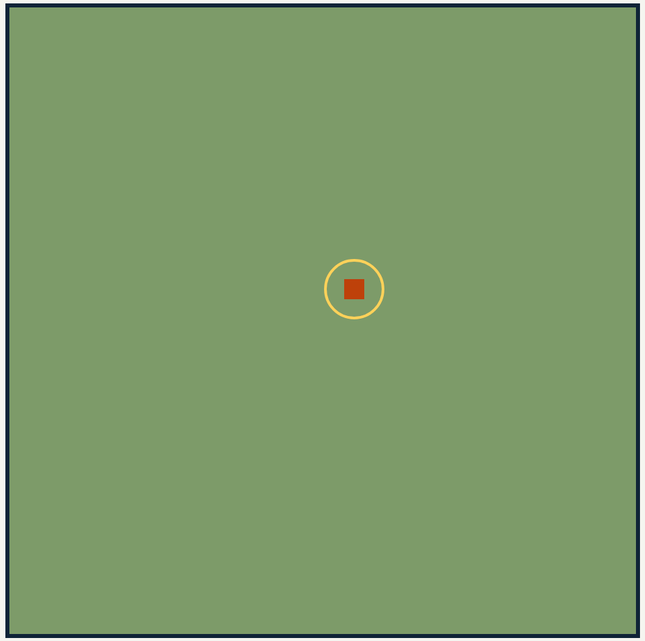
About 32 × 32 m: 354 K against 302 K: easily detected.

Detection tests compare a pixel with its neighbours and the two channels (Giglio et al.).

ACTIVE FIRES

# Grow a fire inside one pixel

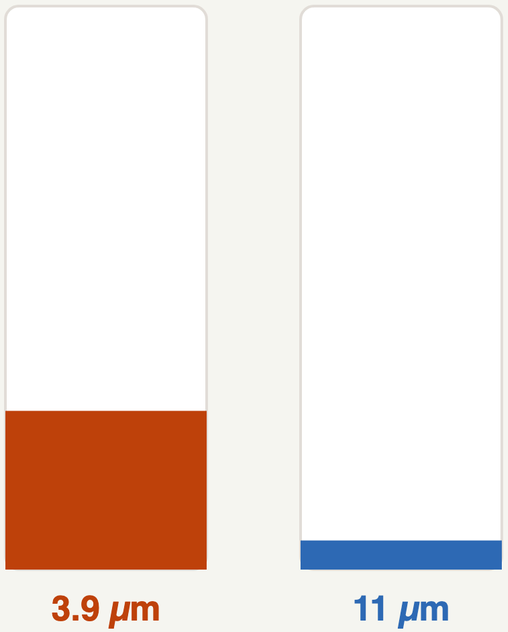
**One 1 km × 1 km pixel**



**Fire: 1,011 m² (32 × 32 m)**

0.10% of the pixel

**Pixel brightness temperature**




**Difference**

**53 K**

**Flagged as fire**

**Simplified test:**

3.9 $\mu$m above 320 K, and
at least 10 K hotter than 11 $\mu$m

Fire 1000 K, ground 300 K

ACTIVE FIRES

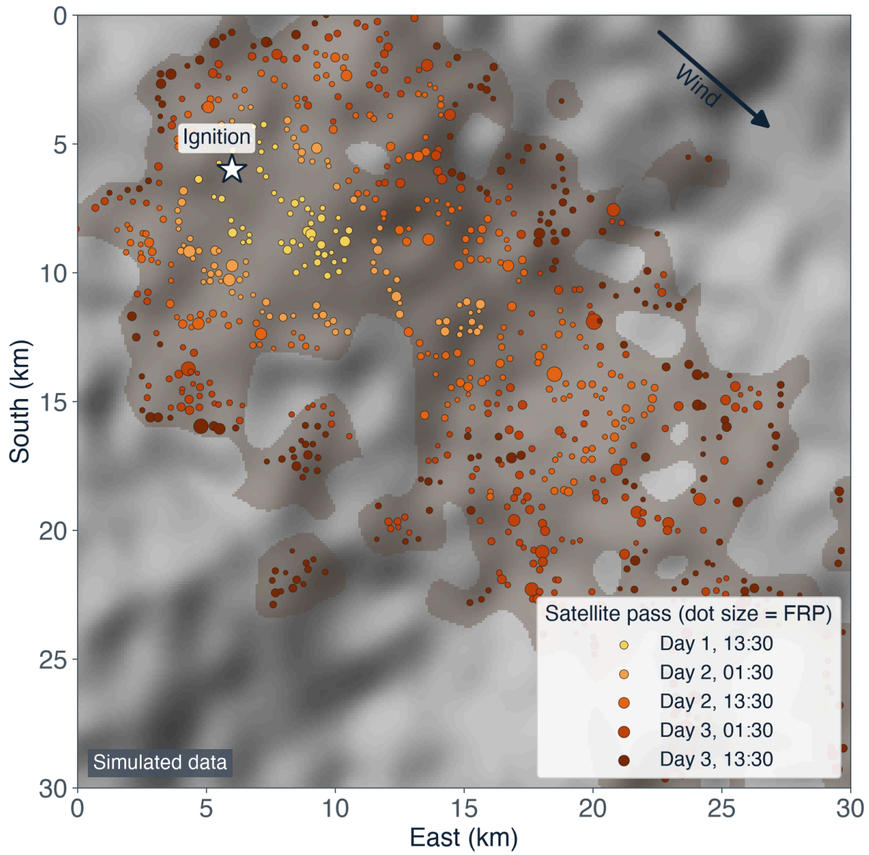
# Satellites that watch fires

| Sensor | Pixel | Strength |
|---|---|---|
| MODIS (Terra, Aqua) | 1 km | 4 passes a day together; record since 2000 |
| VIIRS (Suomi NPP, NOAA-20, NOAA-21) | 375 m | Sees smaller fires; since 2012 |
| Geostationary (GOES, Himawari, Meteosat) | 2–3 km | Every 5–15 minutes: fire growth, timing |
| Sentinel-3 SLSTR | 1 km | Fire radiative power, day and night |
| Sentinel-2, Landsat | 10–30 m | Burn scars and severity; fire fronts in SWIR |

Polar orbiters see small fires in detail a few times a day; geostationary satellites see big fires all day long.

ACTIVE FIRES

# Following a fire from pass to pass




Simulated: VIIRS-like detections, two passes a day.

- Each dot is a 375 m pixel flagged as burning during that pass.
- Successive passes trace the fire front as it moves with the wind.
- Dot size is fire radiative power: how intensely that pixel is burning.
- The shaded area is all the land that burned in three days.

ACTIVE FIRES

# Fire radiative power: how much is burning

$$FRP = \sigma \times A \times (T^4 - T_0^4)$$

Radiant energy given off per second, in megawatts, from the fire's area A and temperature T (Stefan–Boltzmann).

$$\text{Biomass burned} \approx 0.368\ \text{kg} \times \text{FRE (MJ)}$$

Add FRP up over time to get fire radiative energy (FRE), then convert (Wooster et al., 2005).

**Why it matters**

FRP turns a dot into a quantity: fuel consumed, smoke and greenhouse-gas emissions, fire intensity.

**Used by**

Copernicus atmosphere service (smoke forecasts), emission inventories, fire managers.

03

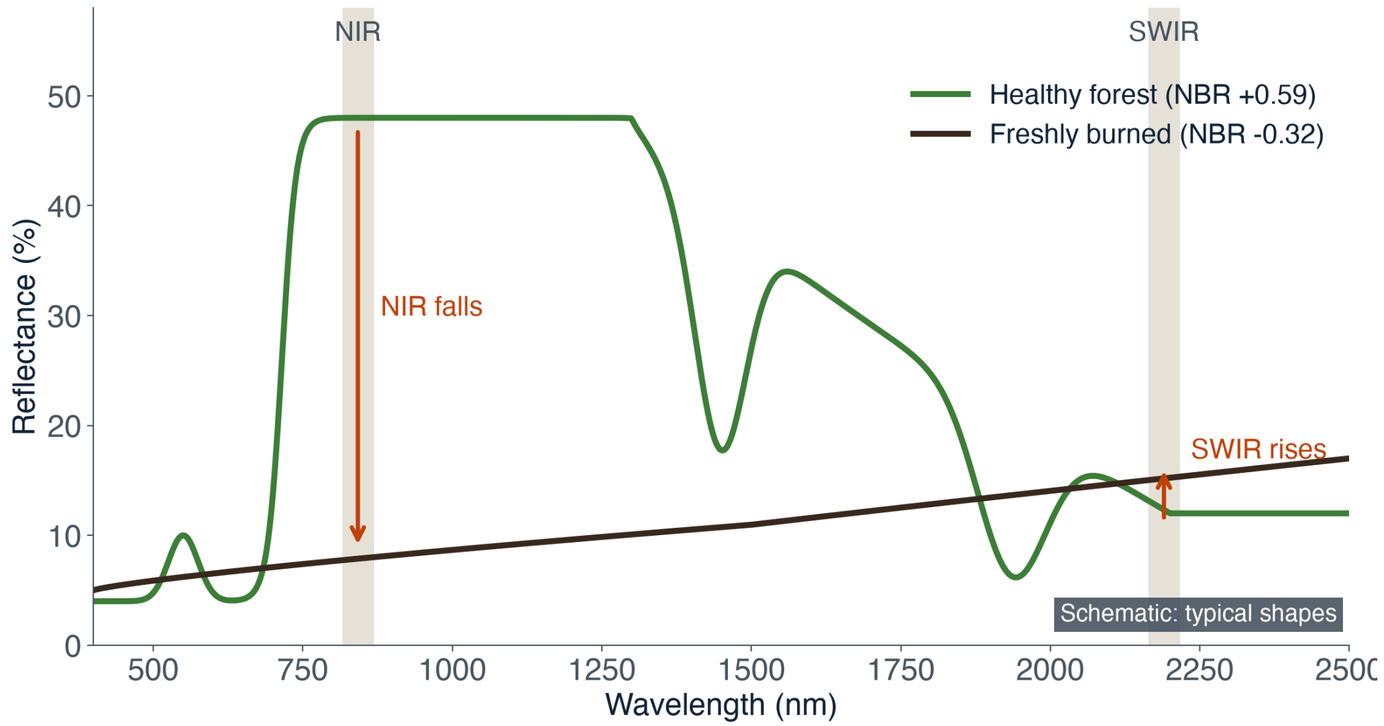
PART 3

# Mapping burned areas and what follows

NBR, burn severity, radar, recovery and post-fire hazards

BURNED AREAS

# Burned land has a different spectrum




Schematic: typical spectra of forest and burned land.

**NBR = (NIR − SWIR) ÷ (NIR + SWIR)**

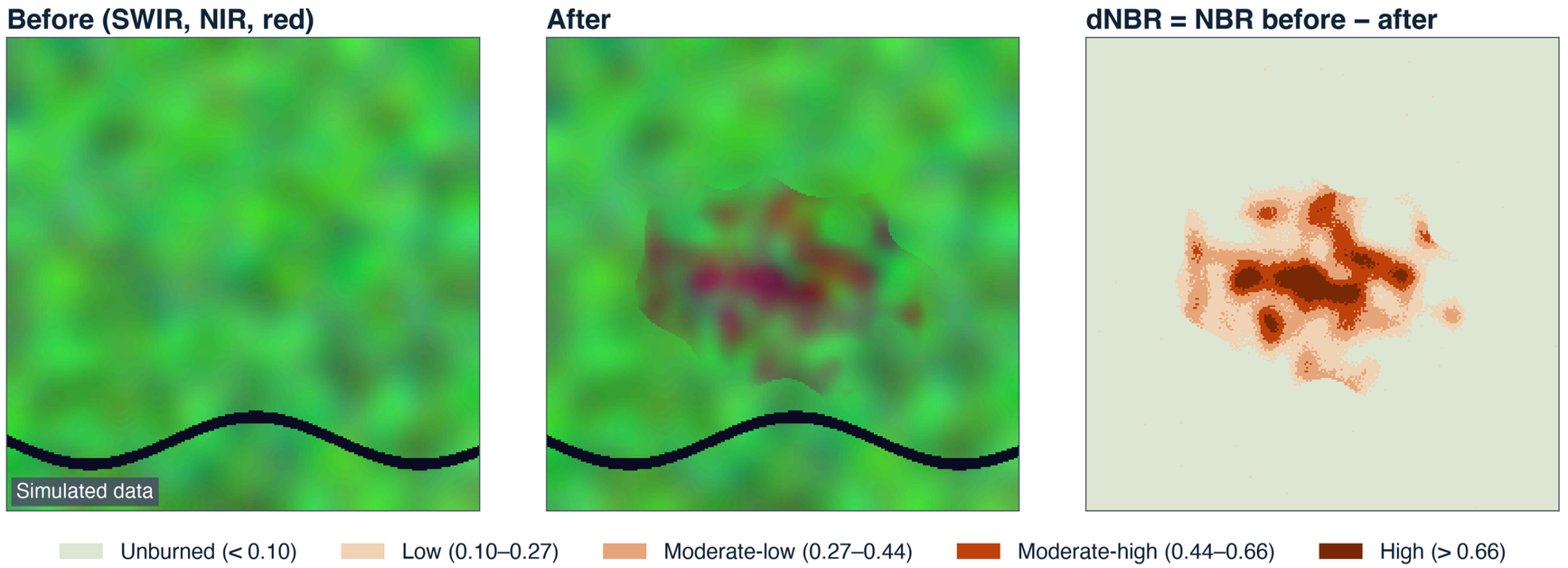
Leaves reflect near-infrared strongly; charcoal and ash do not.

Burning removes leaf water, so short-wave infrared reflectance rises.

Healthy forest ≈ +0.59; freshly burned ≈ −0.32.

BURNED AREAS

# Burn severity from before and after




Simulated scene. Within the scar: 6% high, 13% moderate-high, 22% moderate-low, 28% low severity.

**dNBR:** the bigger the drop in NBR, the more severe the burn.

**Classes:** Key and Benson (2006); calibrate locally with field plots.

BURNED AREAS

# Radar sees through the smoke

**Through smoke and cloud**

Smoke and monsoon cloud hide burn scars from optical sensors for weeks; SAR images every few days regardless.

**Coherence**

Burned forest changes between images, so InSAR coherence falls: a burn signal like the damage maps of Lecture 5.

**Backscatter change**

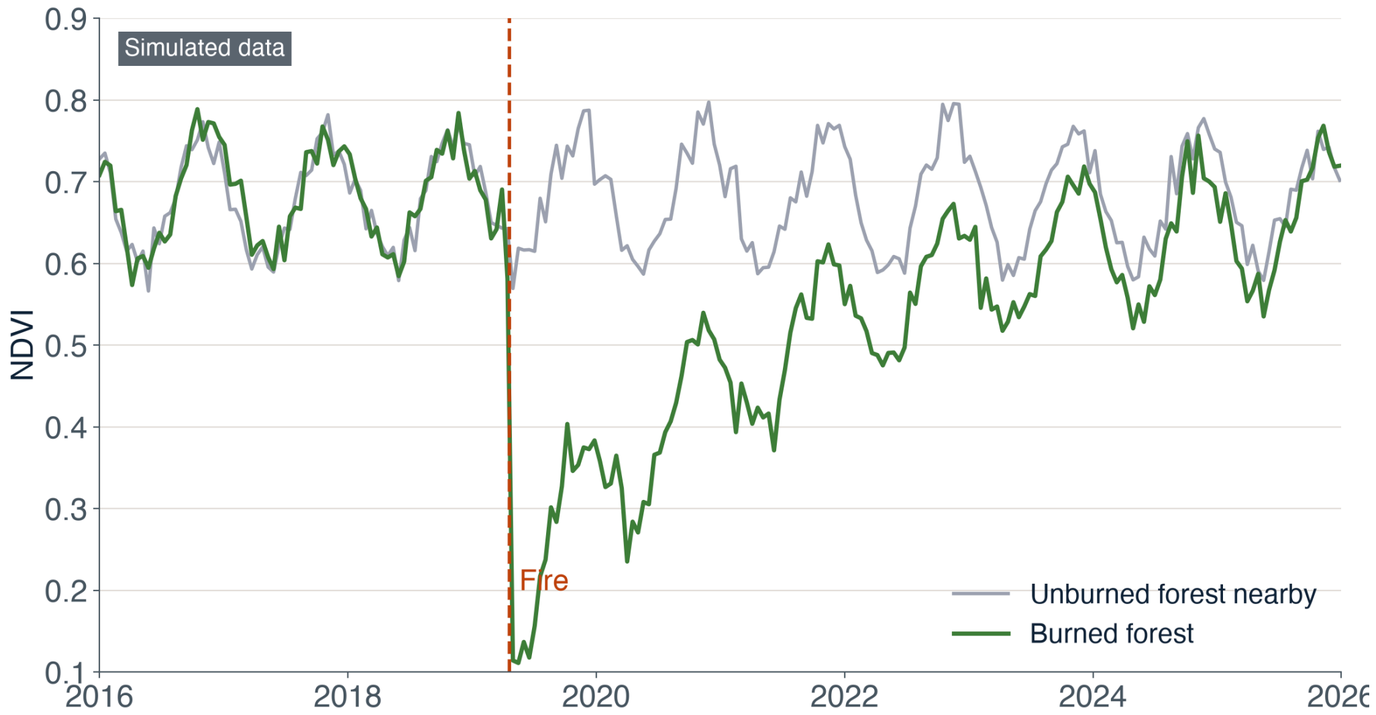
Burned canopy loses leaves and branches: backscatter usually drops, especially in cross-polarisation (VH).

**Best combined**

Use SAR to map quickly during the smoke; refine severity with optical images when the sky clears.

BURNED AREAS

# Watching the forest recover




Simulated: 16-day NDVI composites.

A year after the fire, NDVI is still about 0.32 below the unburned forest.

After five years the gap is only about 0.05: the canopy has largely returned.

Green is not the same as recovered: grass and shrubs can green up long before trees return.

BURNED AREAS

# After the fire: new hazards

**Debris flows**

Burned, water-repellent soil sheds rain fast. The first heavy storms can turn slopes into debris flows.

**Floods**

With vegetation and litter gone, rivers rise faster and higher for several years.

**Landslides**

As burned roots decay over a few years, slopes lose strength (Lecture 4).

**Water supply**

Ash and sediment wash into reservoirs and treatment plants.

04

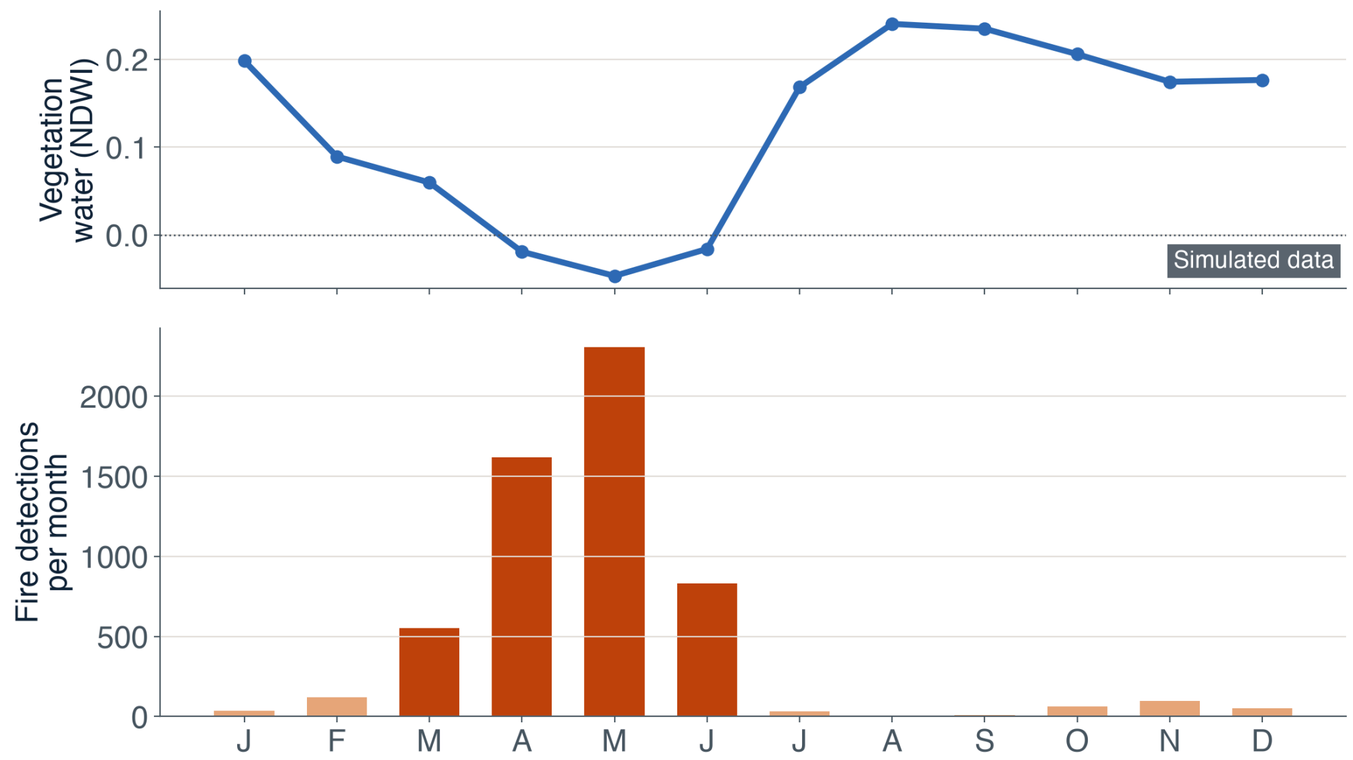
PART 4

# Before the fire: fire danger

Dry vegetation, fire seasons and danger ratings

FIRE DANGER

# Fires follow dryness




Simulated hill-forest district, average year.

Vegetation water indices (such as NDWI) fall through the dry season.

Fire detections peak in the driest weeks, just before the monsoon.

The monsoon wets the fuel and fires almost stop.

The same satellite indices as for drought (Lecture 8) track fuel dryness.

FIRE DANGER

# Fire danger ratings

### Fire Weather Index

The Canadian system: daily temperature, humidity, wind and rain give moisture codes for fuel layers and a spread and intensity index.

### Used worldwide

Copernicus and NASA publish global fire-danger forecasts days ahead using it.

### Satellite inputs

Fuel moisture from vegetation indices and land temperature; fuel type from land-cover maps.

### India

The Forest Survey of India sends weekly pre-fire alerts for areas of very high and extreme danger, based on the Fire Weather Index.

05

PART 5

# Case studies

Uttarakhand, Punjab, Australia, Canada and Los Angeles

CASE STUDY

# Uttarakhand, India, 2023–24 fire season

**Nov–Jun**
Fire season, peaking in April and May 2024

**> 1,000 ha**
Forest burned by early May 2024, by state counts

**900+**
Fire incidents recorded by early May

**2004**
Year FSI began satellite fire alerts (MODIS); VIIRS from 2017

- Surface fires in chir pine forests: dry needles carry fire across the slopes.
- Satellite alerts guided forest crews to remote fires.
- Rain in early May 2024 finally ended the worst of the season.

Sources: Uttarakhand forest department figures reported by India Water Portal and Newslaundry (May 2024); Forest Survey of India.

CASE STUDY

# Crop-residue burning, Punjab

**36,663**
Farm fires counted by satellite in Punjab in 2023

**10,909**
Counted in 2024: about 70% fewer

**Oct–Nov**
Paddy residue burning season

**10:30–13:30**
When the polar satellites pass over

Counts come from satellite hot spots, used to enforce bans.

Many fires moved to late afternoon, after the satellites had passed.

A fall in counts may not mean the same fall in burned area or smoke.

Sources: Punjab government counts (CREAMS, satellite based) as reported in the press; CEEW (2024).

CASE STUDY

# Black Summer, Australia, 2019–20

**24 million ha**
Burned across Australia

**33**
People killed directly by the fires

**400+**
Further deaths linked to smoke

**3,000+**
Homes destroyed

- Record drought and heat dried forests that rarely burn.
- Smoke reached the stratosphere and circled the Southern Hemisphere.
- Satellites mapped fire spread daily, and burn severity afterwards.

Sources: CSIRO; Australian Institute for Disaster Resilience; Filkov et al. (2020).

CASE STUDY

# Canada, 2023

**15 million ha**
Burned: more than double the previous record

**2.5 million ha**
A typical year

**6,000+**
Fires during the season

**≈ 232,000**
People evacuated

Smoke spread across North America and reached Europe.

Satellites followed thousands of fires at once: no ground network could.

Climate change made a season this large far more likely.

Sources: Natural Resources Canada; Jain et al. (2024), Nature Communications; Kirchmeier-Young et al. (2024), npj Climate and Atmospheric Science.

CASE STUDY

# Los Angeles, January 2025

**16,000+**
Structures destroyed by the Palisades and Eaton fires

**≈ 23,000 acres**
Burned by the Palisades fire alone

**Santa Ana**
Strong, dry winds drove the fires into towns

**At least 31**
People killed in the January 2025 fires

- Wind-driven fires spread into the wildland–urban interface faster than crews could respond.
- Satellites tracked smoke and heat, and mapped damage afterwards.
- Engineering lessons: defensible space, ember-resistant buildings, water supply.

Sources: CAL FIRE; news reports (January 2025).

06

PART 6

# Engineering practice

Uses, a workflow, limits and exercises

ENGINEERING

# Where engineers use fire data

**Wildland–urban interface**

Zoning, setbacks and fire-resistant building near forests.

**Lifelines**

Power lines, pipelines and roads through forests: clearance and shut-off planning.

**Post-fire hazards**

Debris-flow and flood warnings, check dams, culvert sizing.

**Water supply**

Protect reservoirs and catchments from ash and sediment.

**Air quality**

Smoke forecasts for cities and health advice.

**Fire management**

Fire lines, patrols and controlled burning where danger is high.

ENGINEERING

# A wildfire monitoring workflow

01
Fire-danger map (before)

02
FIRMS alerts (during)

03
Check and dispatch crews

04
Track spread and FRP

05
Map burn scar: dNBR

06
Classify severity

07
Post-fire hazards

08
Monitor recovery

**Free tools:** NASA FIRMS for alerts; Google Earth Engine for NBR from Sentinel-2 and Landsat.

**Student project:** map one fire season in your state, and its burn severity.

ENGINEERING

# Limits and good practice

### Limits

- Small, short or cool fires are missed
- Fires under cloud, thick smoke or forest canopy
- Fixed overpass times miss fires in between
- Hot pixels are not exact fire locations

### Good practice

- Combine polar and geostationary satellites
- Check alerts on the ground
- Use burned-area maps, not only hot-spot counts
- Calibrate burn severity with field plots

EXERCISES

# Try these

**01** Using Wien’s law ($\lambda = 2898 \div T$ μm), at what wavelength do ground at 300 K and a fire at 1000 K glow most? Why use a 3.9 μm channel?

**02** Before a fire, NIR = 0.40 and SWIR = 0.10. After, NIR = 0.15 and SWIR = 0.20. Find NBR before and after, dNBR, and the severity class.

**03** A fire covers 1,000 m² at 1000 K on 300 K ground. What is its FRP? ($\sigma = 5.67 \times 10^{-8}$ W m$^{-2}$ K$^{-4}$)

**04** If that fire burns steadily for 10 hours, about how much biomass does it consume? (0.368 kg per MJ)

EXERCISES

# Answers

**01** 300 K: about **9.7 µm**; 1000 K: about **2.9 µm**. Near 3–4 µm the fire outshines the ground thousands of times, so even a small fire shows.

**02** Before (0.40 − 0.10) ÷ 0.50 = **0.60**; after (0.15 − 0.20) ÷ 0.35 ≈ **−0.14**; dNBR ≈ **0.74**: high severity.

**03** $5.67 \times 10^{-8} \times 1{,}000 \times (1000^4 - 300^4) \approx$ **56 MW**.

**04** FRE = 56 MW × 36,000 s ≈ $2.0 \times 10^6$ MJ; × 0.368 kg/MJ ≈ **745 tonnes** of biomass.

RECAP

# Key points

01 Fires glow strongly in the middle infrared; satellites compare it with thermal infrared.

02 A fire far smaller than a pixel can still be detected.

03 FRP measures how intensely it burns; it converts to fuel burned and smoke.

04 NBR and dNBR map burn scars and severity; radar works through smoke.

05 Use danger ratings before, alerts during, and severity maps after a fire.

FURTHER READING

# References


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NASA FIRMS: Fire Information for Resource Management System, firms.modaps.eosdis.nasa.gov.

# Thank you. Questions?

Figures and the animation were made for this lecture. The Planck curves and sub-pixel temperatures are computed from physics; the fire detections, burn-severity scene, recovery series and fire-season chart are simulated, and labelled as such; the burned-land spectra are schematic. Severity classes after Key and Benson (2006); biomass factor after Wooster et al. (2005). Case-study facts: Uttarakhand forest department and Forest Survey of India; Punjab government and CEEW; CSIRO and AIDR (Australia); Natural Resources Canada; CAL FIRE (Los Angeles). Full references on the previous slide.